

The Alpha-Beta Bounds: Branching, Particle Ordering, and Search Determinism

Theorem monograph T10 of the chess-dynamics-lab program —
the full game

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Keywords: alpha-beta, negamax, branching, move ordering, determinism

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Problem statement

The full game is the second loop of the engine: the search minimizes the Lagrangian energy of T05 over the tree of legal moves. Naive minimax enumerates b^d leaves, where b is the average branching factor (for chess $b \approx 30-35$), and already at $d = 6$ demands billions of nodes. Alpha-beta prunes branches that provably cannot affect the choice of the root move; the scale of the gain — from zero to the square root of the exponent — depends solely on the order in which moves are examined.

In the particle model the move order is kinematics: moves of the particles with the largest expected energy change are examined first. The practical carrier of the idea is the MVV-LVA heuristic (most valuable victim — least valuable aggressor) with a kinetic centralization bonus: the capture of an expensive piece by a cheap particle is the most likely source of a sharp energy drop, so such moves come first and produce cutoffs earlier. The ordering does not change the root value — only the path to it and the cost of the path.

The third requirement is determinism: the same input must produce the same move, the same node count and the same principal variation. For a certifiable laboratory this is not a matter of style but a condition of protocol reproducibility: comparing node counts and lines across runs, machines and languages is meaningful only under a fully deterministic search.

Theorem

Theorem T10 (alpha-beta correctness, bounds and determinism). Let T_d be the tree of legal continuations of depth d from position P , b the branching

factor. Then: (i) negamax with the window (alpha, beta) returns the same value and the same best move as full minimax; (ii) in the worst move order the number of visited leaves is b^d , in the best — $b^{\text{ceil}(d/2)} + b^{\text{floor}(d/2)} - 1$; (iii) with MVV-LVA and centralization ordering on the start position the search visits: $d = 2$ — 79 nodes against 421 of the full tree, $d = 3$ — 731 against 9323, $d = 4$ — 3345 against 206604, $d = 5$ — 19753 against 5072213 (gains 5.3x, 12.8x, 61.8x, 256.8x respectively); (iv) under a total order on the move set the search is deterministic: the move, the value and the node count are bit-reproducible.

Proof

(i) Correctness. The negamax identity $\text{val}(P) = \max_m [-\text{val}(P \cdot m)]$ reduces minimax to a single form for both sides: the score is always from the point of view of the side to move, and the parent maximizes the negated child. Induction over d : a beta-cutoff returns the bound beta rather than the exact value, but by the inductive hypothesis the exact value of the pruned subtree lies outside the window required by the parent and therefore can change neither the root value nor the argument of the maximum. This is the classical Knuth-Moore argument (1975) in negamax form; the laboratory additions — quiescence at the leaves (searching only captures until the position calms down) and the mate encoding $-\text{MATE} + \text{ply}$, which prefers the shorter mate — do not alter the argument since they preserve the monotonicity of the score over the tree. The 50-move rule ($\text{counter} \geq 100$) returns zero — a draw.

(ii) Bounds. The worst order: no cutoff ever fires, the tree is traversed entirely — b^d leaves. The best order (the cutoff move always first): after the first root move the window collapses, and every subsequent subtree requires a full expansion of only half of its depth; the Knuth-Moore count gives $b^{\text{ceil}(d/2)} + b^{\text{floor}(d/2)} - 1$ leaves — the square root of the exponent instead of the exponent itself. Between the bounds the quality of the ordering heuristic decides everything; MVV-LVA is the standard chess compromise between the accuracy of cutoff prediction and the cost of sorting.

(iii) Measurements. The node counter counts negamax calls, including quiescence nodes; the full-tree reference is the sum of $\text{perft}(0..d)$ with the perft values certified in T08. The gain grows with depth, approaching the prediction of (ii): this agrees with the quality of MVV-LVA in chess openings, where captures really do concentrate near the top of the tree of variations. The values are pinned by the protocol as reference: any regression of generation or ordering changes them and is caught by comparison with the frozen baseline.

(iv) Determinism. `order_moves` sorts the moves by a stable sort over a total numeric key; the input order is determined by the fixed 0x88 board traversal of the generator; equal keys keep the generation order. All search operations are pure: `make/unmake` restore the state exactly (verified in T09). Hence the search trajectory is a function of the pair (position, depth), and identical runs produce bit-identical results on any machine and in any language. The theorem is proved.

Protocol data

Alpha-beta nodes on the start position (MVV-LVA + centralization)

Depth	Search nodes	Full tree	Gain
2	79	421	× 5.3
3	731	9323	× 12.8
4	3345	206604	× 61.8
5	19753	5072213	× 256.8

Search service: components affecting the bounds

Component	Role	Certificate
MVV-LVA + centralization	move order, early cutoffs	node table (iii)
quiescence	captures at leaves, horizon removal	selfplay self-consistency
mate code –MATE + ply	preference of the shortest mate	tactical reference T11(iv)
50-move rule	draw at halfmove ≥ 100	pytest tests
stable sort	total move order	reproducibility (iv)

Summary of what is proved

We proved the correctness of the windowed negamax search, reproduced the classical worst- and best-case bounds, pinned the protocol node measurements (up to 256.8x at depth 5) and proved the determinism of the search. The full-game loop thereby rests on the same certified foundations as the retrograde analysis: the generation of T08, the memory of T09 and the energy of T05.

Verification

python3 dynamics.py -analyze <FEN> -depth 4 (move, score, nodes); python3 engine/game_player.py -selfplay -depth 4 (deterministic self-play); the forced checks of C8 run on the same search core. The comparison of node counts with the table of the theorem is performed by the pytest tests (baseline_c1_c9.json).